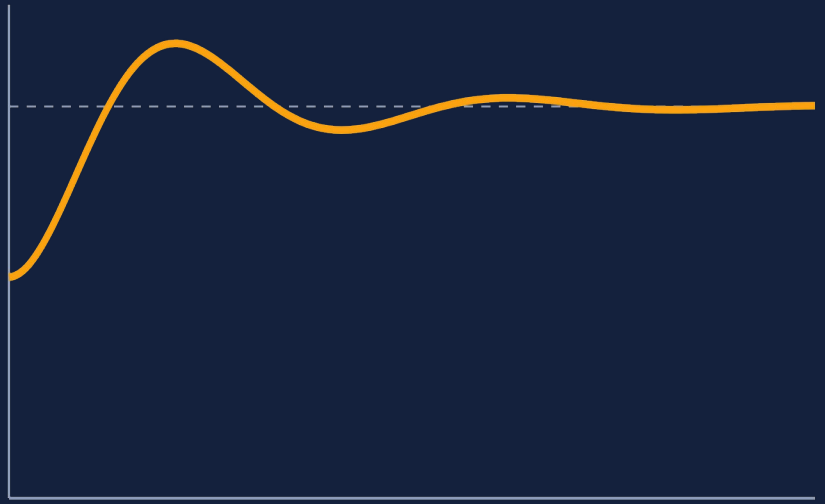
Control Systems and Transfer Functions



Control Systems (22EEG53)

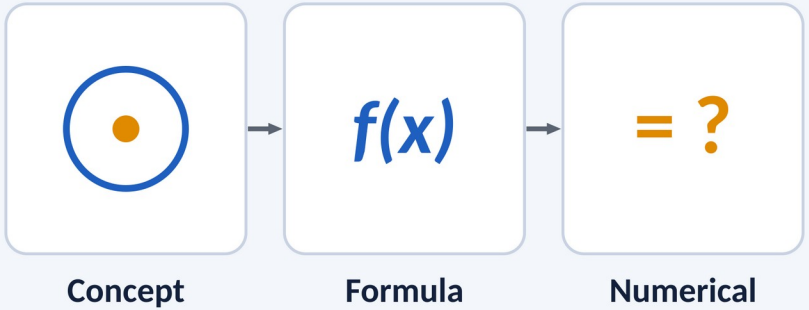
Unit 1: Mathematical Modelling of Physical Systems

Topic: Open/closed loop systems, components, transfer functions of mechanical and electrical systems

Prof. Mitesh Astik

Learning Outcomes

- Distinguish open-loop and closed-loop systems with examples.
- Name the components of a feedback control system.
- Write transfer functions of mechanical and electrical systems.
- Use the force–voltage analogy between the two.



Linked to CO1
Model dynamical systems by transfer functions; deduce mechanical–electrical analogy.

Recap: What You Already Know

- Laplace transform turns differential equations into algebra.
- Newton's law links force, mass and acceleration.
- Kirchhoff's laws give circuit equations.
- Poles are roots of the denominator.

Laplace transform
$\mathcal{L}\{f'(t)\} = sF(s) - f(0)$

Newton's second law
$\Sigma F = M \cdot a$

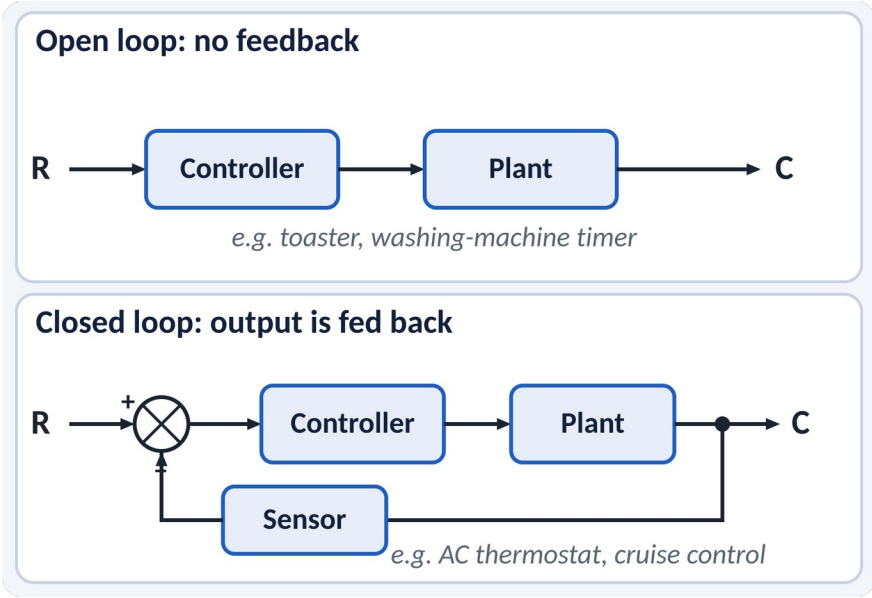
Kirchhoff's voltage law
$\Sigma v = 0$ around a loop

Partial fractions
factor $D(s)$ into poles

Zero initial conditions: $\mathcal{L}\{dx/dt\} = sX(s)$, $\mathcal{L}\{d^2x/dt^2\} = s^2X(s)$
$X(s)$ Laplace transform of $x(t)$, s complex frequency (1/s)

Motivation: Why Feedback?

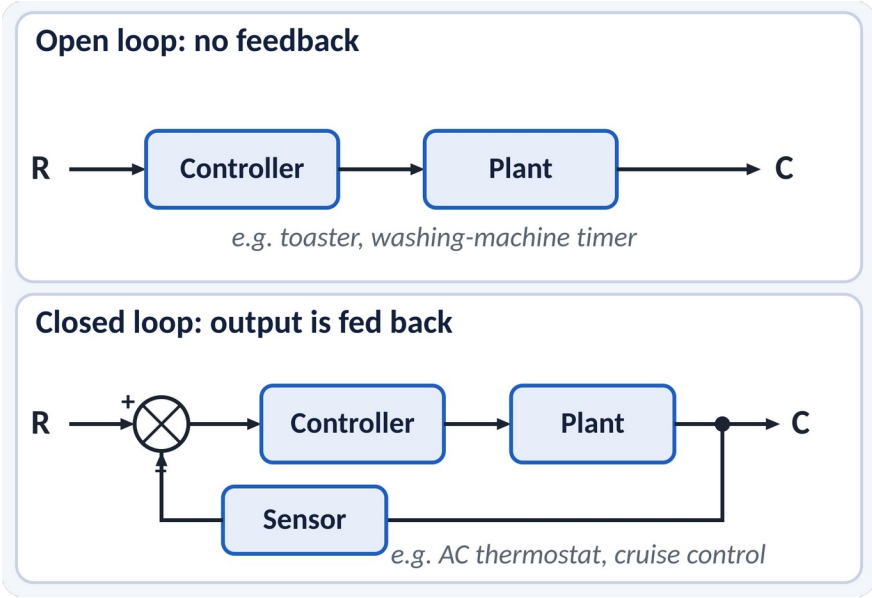
- Cruise control keeps speed on a hill.
- Home AC keeps room temperature near set value.
- Robot arm must reach the commanded position.
- Feedback corrects errors caused by disturbances.



Control aim: make the output follow the reference despite disturbances

Open Loop and Closed Loop

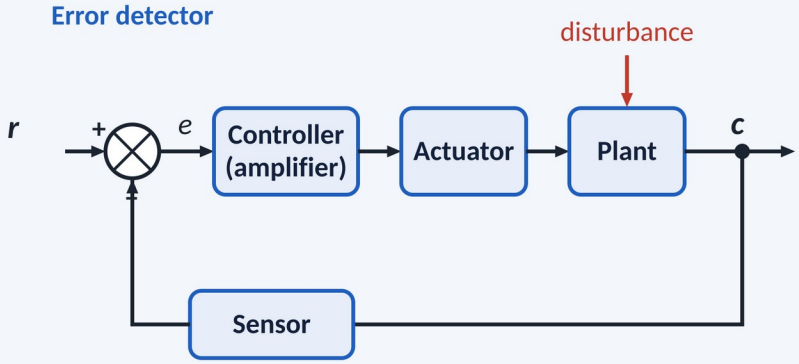
- Open loop: output has no effect on control action.
- Closed loop: output is measured and fed back.
- Open loop: simple, cheap, but sensitive to disturbance.
- Closed loop: accurate, but more complex; can become unstable.



Closed loop: $e = r - b$, $b = H \cdot c$
r reference, c output, b feedback signal, e error

Components of a Control System

- Error detector: compares reference with feedback.
- Controller or amplifier: processes the error signal.
- Actuator: motor or valve that drives the plant.
- Plant: the process being controlled.
- Sensor: measures output, e.g. tachometer or synchro.

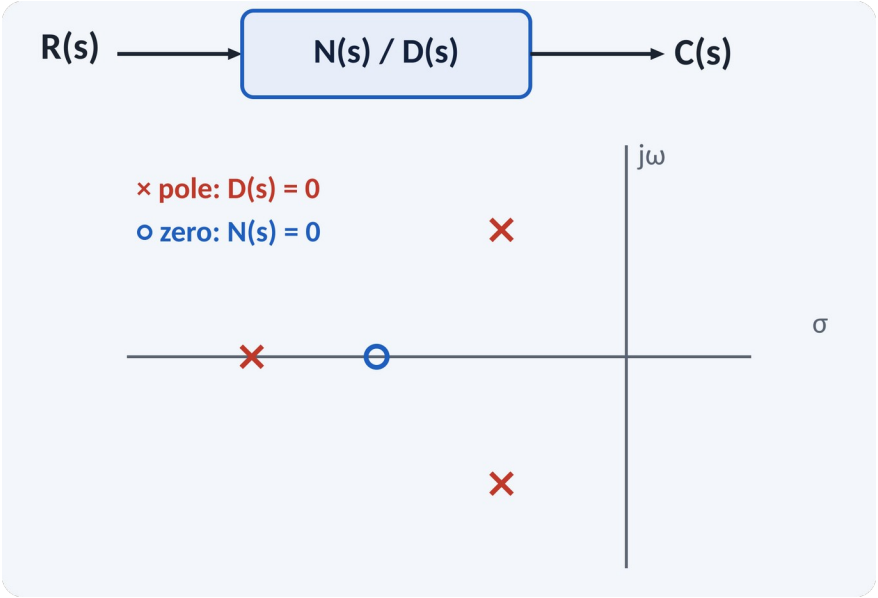


Disturbance acts on the plant; the sensor measures the output

Examples: potentiometer, synchro pair, tachogenerator, DC servo motor

Transfer Function

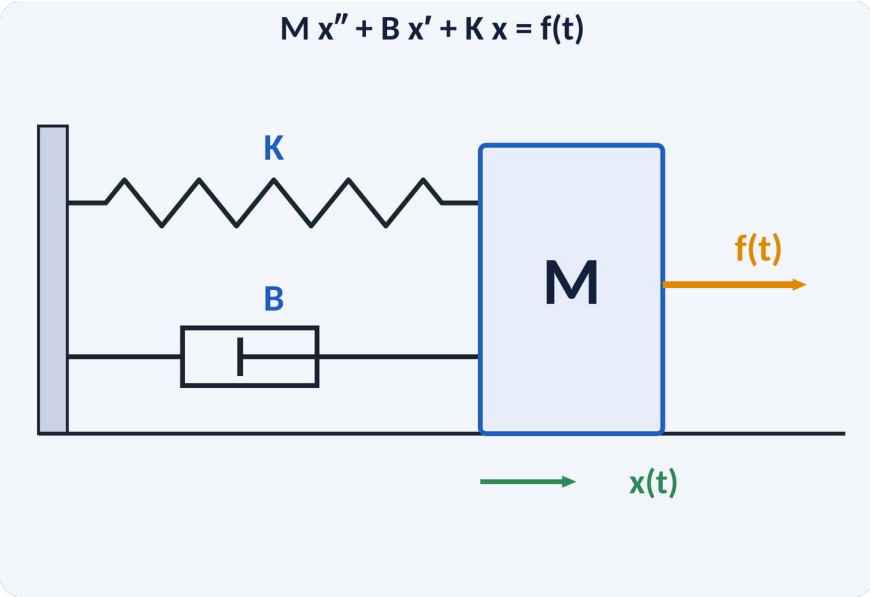
- Ratio of output to input in Laplace domain.
- Initial conditions are taken as zero.
- Valid for linear time-invariant systems.
- Roots of D(s) are poles; roots of N(s) are zeros.



G(s) = C(s)/R(s) = N(s)/D(s)
D(s) = 0 is the characteristic equation of the system

Translational Mechanical Systems

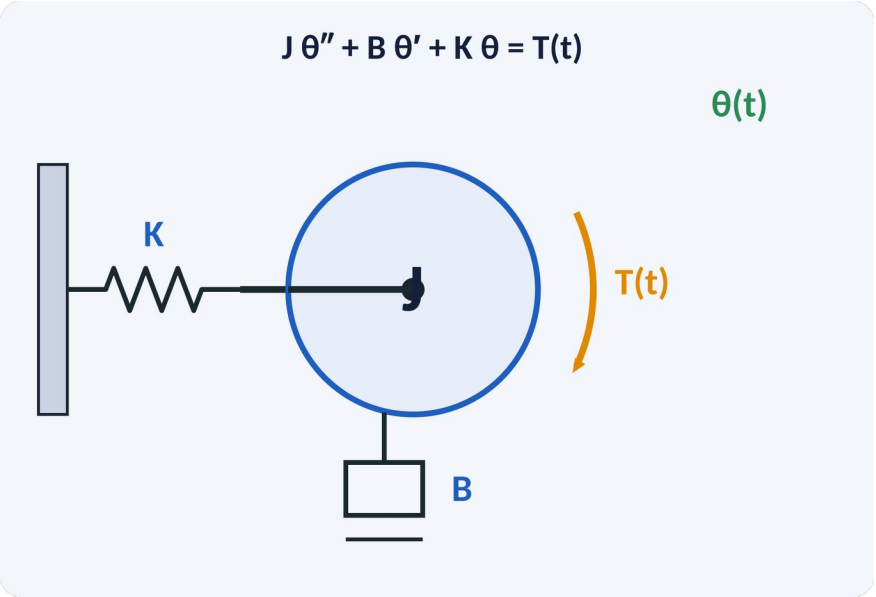
- Mass M: opposes acceleration, force = M·x″.
- Damper B: opposes velocity, force = B·x′.
- Spring K: opposes displacement, force = K·x.
- Apply Newton's law to the mass.
- Take the Laplace transform and divide.



$M\,x'' + B\,x' + K\,x = f(t) \;\rightarrow\; X(s)/F(s) = 1/(M\,s^2 + B\,s + K)$
M kg, B N·s/m, K N/m, x m, f N

Rotational Mechanical Systems

- Inertia J: torque = J·θ″.
- Viscous friction B: torque = B·θ′.
- Torsional spring K: torque = K·θ.
- Same steps: torque balance, Laplace, divide.

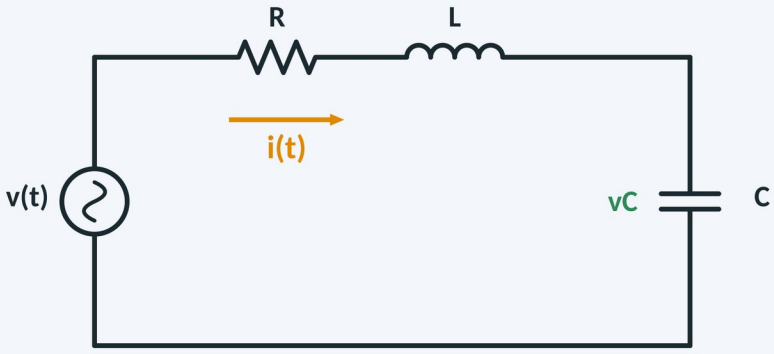


$J\,\theta'' + B\,\theta' + K\,\theta = T(t) \;\rightarrow\; \Theta(s)/T(s) = 1/(J\,s^2 + B\,s + K)$

J kg·m², B N·m·s/rad, K N·m/rad, θ rad, T N·m

Electrical Systems

- Resistor: v = R·i.
- Inductor: v = L·di/dt.
- Capacitor: v = (1/C)∫i dt.
- Apply KVL, then Laplace transform.
- Choose the output, e.g. voltage across C.



$$v = R\cdot i + L\cdot di/dt + (1/C)\int i\,dt$$

$VC(s)/V(s) = 1/(L\,C\,s^2 + R\,C\,s + 1)$
R Ω, L H, C F, i A, v V

Analogous Systems

- Same differential equation, different physical quantities.
- Force–voltage analogy: M ↔ L, B ↔ R, K ↔ 1/C.
- Force–current analogy: M ↔ C, B ↔ 1/R, K ↔ 1/L.
- Lets us study a mechanical system with a circuit.

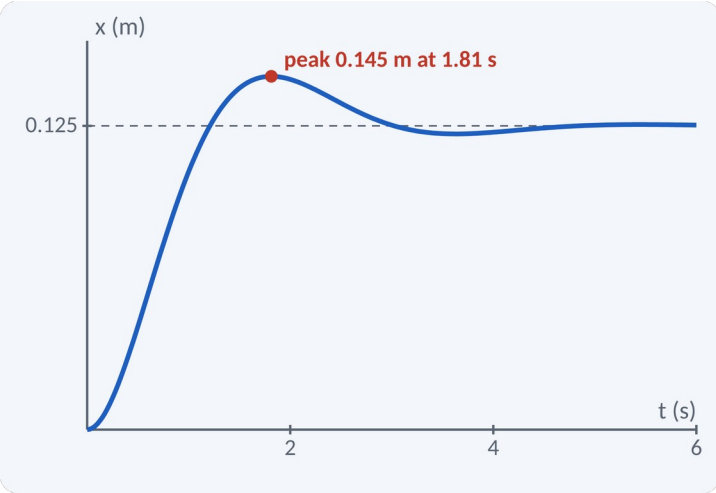
Mechanical (force–voltage analogy)		Electrical
Force f	↔	Voltage v
Mass M	↔	Inductance L
Damper B	↔	Resistance R
Spring K	↔	1/C (elastance)
Displacement x	↔	Charge q
Velocity $\dot{x}$	↔	Current i

$$M x'' + B x' + K x = f \iff L q'' + R q' + (1/C) q = v$$

$x \leftrightarrow q, \dot{x} \leftrightarrow i$

Solved Example 1: Mass–Spring–Damper

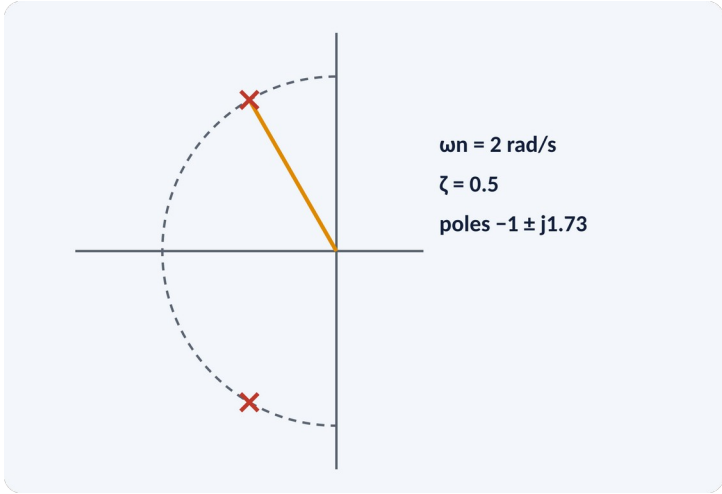
- Given M = 2 kg, B = 4 N·s/m, K = 8 N/m. Find X(s)/F(s), ωn, ζ and final value for a 1 N step.
- 1. X/F = 1/(2s² + 4s + 8).
- 2. ωn = √(K/M) = √4 = 2 rad/s.
- 3. ζ = B/(2√(KM)) = 4/(2√16) = 0.5.
- 4. Final x = F/K = 1/8 = 0.125 m.
- 5. Mp = e^(−π·0.5/√0.75) = 16.3 %; peak = 0.125 × 1.163 = 0.145 m.



Answer: ωn = 2 rad/s, ζ = 0.5 (underdamped), x(∞) = 0.125 m

Solved Example 2: Series RLC and Analogy

- Given R = 2 Ω, L = 1 H, C = 0.25 F. Find VC/V, ωn, ζ and the mechanical analogue.
- 1. VC/V = $1/(0.25s^2 + 0.5s + 1) = 4/(s^2 + 2s + 4)$.
- 2. ωn = 2 rad/s.
- 3. 2ζωn = 2, so ζ = 0.5.
- 4. Poles: s = −1 ± j1.73.
- 5. Analogue: M = L = 1 kg, B = R = 2 N·s/m, K = 1/C = 4 N/m.



Answer: ωn = 2 rad/s, ζ = 0.5; analogous M = 1 kg, B = 2, K = 4

Common Mistakes and Exam Tips

Common mistakes

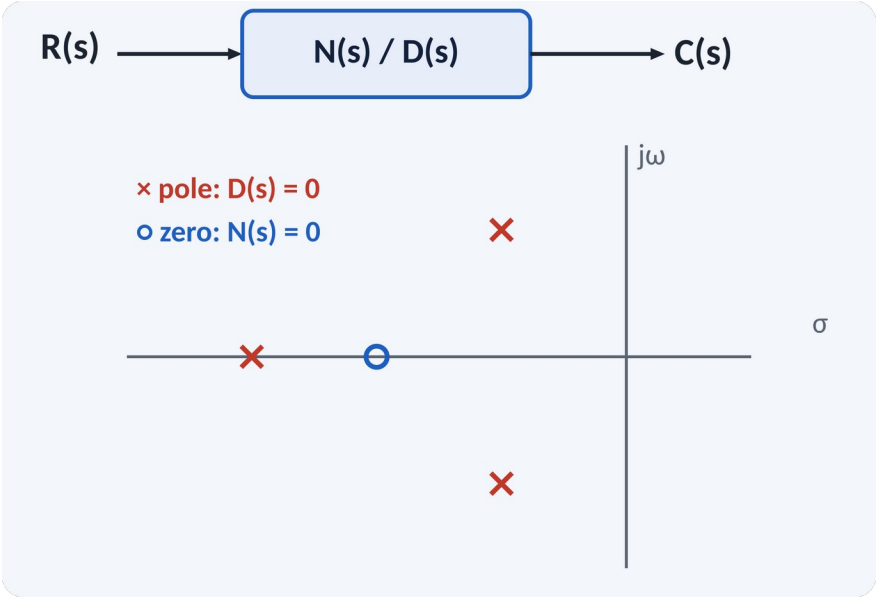
- Forgetting zero initial conditions in G(s).
- Using the wrong sign for the feedback summing junction.
- Mixing up the force–voltage and force–current analogies.
- Using grams or cm instead of SI units.
- Treating a nonlinear system as if G(s) exists.

Exam tips

- Draw the free body diagram first.
- Write the equation, then Laplace transform.
- Compare with $1/(M\,s^2 + B\,s + K)$ to check.
- State units with every parameter.

Summary

- Closed loop feeds the output back to reduce error.
- Components: error detector, controller, actuator, plant, sensor.
- G(s) = C(s)/R(s) with zero initial conditions.
- Mechanical: $1/(M s^2 + B s + K)$; same form for J, B, K.
- Series RLC: $1/(L C s^2 + R C s + 1)$; M ↔ L, B ↔ R, K ↔ 1/C.



G(s) = N(s)/D(s): poles from D(s), zeros from N(s)

Quick Quiz

Q1. A thermostat-controlled room heater is:
(a) open loop (b) closed loop (c) nonlinear only (d) digital only

Ans: (b)

Q2. X(s)/F(s) for M, B, K system is:
(a) $1/(Ms^2+Bs+K)$ (b) Ms^2+Bs+K (c) $1/(Ms+B)$ (d) $K/(Ms^2+B)$

Ans: (a)

Q3. In the force–voltage analogy, mass corresponds to:
(a) resistance (b) capacitance (c) inductance (d) charge

Ans: (c)

Q4. Poles of G(s) are the roots of:
(a) numerator (b) denominator (c) input (d) output

Ans: (b)

Q5. For M = 2 kg, K = 8 N/m, ωn is:
(a) 1 rad/s (b) 2 rad/s (c) 4 rad/s (d) 16 rad/s

Ans: (b)

References

1. **I. J. Nagrath and M. Gopal, Control Systems Engineering**
   5th Ed., New Age International (P) Ltd., 2008 (Units 1, 2 textbook)

2. **K. Ogata, Modern Control Engineering**
   5th Ed., Pearson Education Asia / PHI, 2009 (reference)

3. **P. S. Satyanarayana, Concepts of Control Systems**
   2nd Ed., Dynaram Publishers, Bangalore, 2001 (reference)

Next lecture: block diagram reduction and signal flow graphs (Mason's gain formula)